

Solution to MP363 Problem Set 4

(1)

P.1

We have a state in an infinite square well given by

$$\psi(t, x) = \frac{1}{\sqrt{5}} \phi_1(t, x) - \frac{2i}{\sqrt{5}} \phi_3(t, x)$$

where

$$\phi_n(t, x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) e^{-iE_n t/\hbar} & 0 \leq x \leq a \\ 0 & x \leq 0, x \geq a \end{cases}$$

where $E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2$ and $n = 1, 2, 3, \dots$

(a) There's a very easy way to do this, and a less easy way. The easy way is

to realize that the energy eigenstates are orthonormal, i.e. $\langle \phi_n | \phi_m \rangle = \delta_{nm}$. This means that if a state is written as $|\psi\rangle = \sum_n c_n |\phi_n\rangle$, then

$$\langle \psi | \psi \rangle = \sum_n |c_n|^2 \text{ and so the state is normalized if } \sum_n |c_n|^2 = 1.$$

For this state, only ϕ_1 and ϕ_3 appear, so $c_n = 0$ for $n \neq 1$ or 3 .

Thus, $\sum_{n=1}^{\infty} |c_n|^2 = |c_1|^2 + |c_3|^2$. But $c_1 = \frac{1}{\sqrt{5}}$ and $c_3 = -\frac{2i}{\sqrt{5}}$, so

$$|c_1|^2 + |c_3|^2 = \left|\frac{1}{\sqrt{5}}\right|^2 + \left|-\frac{2i}{\sqrt{5}}\right|^2 = \frac{1}{5} + \frac{4}{5} = 1, \text{ and so } \boxed{\text{the state has unit norm.}}$$

The less-easy way is to do the integral: with that,

$$\psi(t, x) = \sqrt{\frac{2}{5a}} \left[\sin\left(\frac{\pi x}{a}\right) e^{-iE_1 t/\hbar} - 2i \sin\left(\frac{3\pi x}{a}\right) e^{-iE_3 t/\hbar} \right].$$

This is the sum of two complex numbers, and since $|z_1 + z_2|^2 = (z_1 + z_2)(z_1^* + z_2^*) = |z_1|^2 + |z_2|^2 + 2\text{Re}(z_1^* z_2)$,

taking $z_1 = \sin\left(\frac{\pi x}{a}\right) e^{-iE_1 t/\hbar}$ and $z_2 = -2i \sin\left(\frac{3\pi x}{a}\right) e^{-iE_3 t/\hbar}$ gives

$$|\psi(t, x)|^2 = \frac{2}{5a} \left\{ \left| \sin\left(\frac{\pi x}{a}\right) e^{-iE_1 t/\hbar} \right|^2 + \left| -2i \sin\left(\frac{3\pi x}{a}\right) e^{-iE_3 t/\hbar} \right|^2 + 2\text{Re}\left(\left(\sin\left(\frac{\pi x}{a}\right) e^{-iE_1 t/\hbar} \right) \left(-2i \sin\left(\frac{3\pi x}{a}\right) e^{-iE_3 t/\hbar} \right)^* \right) \right\}$$

$$= \frac{2}{5a} \left[\sin^2\left(\frac{\pi x}{a}\right) + 4 \sin^2\left(\frac{3\pi x}{a}\right) - 4 \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{3\pi x}{a}\right) \text{Re}\left(i e^{i(E_1 - E_3)t/\hbar} \right) \right].$$

Since

$$e^{i(E_1 - E_3)t/\hbar} = i \left[\cos\left(\frac{E_1 - E_3}{\hbar} t\right) + i \sin\left(\frac{E_1 - E_3}{\hbar} t\right) \right] = \sin\left(\frac{4\pi^2 \hbar^2 t}{ma^2}\right) + i \cos\left(\frac{4\pi^2 \hbar^2 t}{ma^2}\right)$$

(since $E_1 - E_3 = \frac{\pi^2 \hbar^2}{2ma^2} - \frac{9\pi^2 \hbar^2}{2ma^2} = -\frac{4\pi^2 \hbar^2}{2ma^2}$), we find that

$\sin\left(\frac{4\pi^2 \hbar^2 t}{ma^2}\right)$. Thus,

$$|\psi(t, x)|^2 = \frac{2}{5a} \left[\sin^2\left(\frac{\pi x}{a}\right) + 4 \sin^2\left(\frac{3\pi x}{a}\right) - 4 \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{3\pi x}{a}\right) \sin\left(\frac{4\pi^2 \hbar^2 t}{ma^2}\right) \right]$$

So integrating this for $0 \leq x \leq a$ will require three integrals:

$$\int_0^a \sin^2\left(\frac{\pi x}{a}\right) dx = \int_0^a \frac{1}{2} \left[1 - \cos\left(\frac{2\pi x}{a}\right) \right] dx = \frac{1}{2} \left[x - \frac{a}{2\pi} \sin\left(\frac{2\pi x}{a}\right) \right]_0^a = \frac{1}{2} \left[\left(a - \frac{a}{2\pi} \sin(2\pi) \right) - (0 - 0) \right] = \frac{a}{2}$$

$$\int_0^a \sin^2\left(\frac{3\pi x}{a}\right) dx = \int_0^a \frac{1}{2} \left[1 - \cos\left(\frac{6\pi x}{a}\right) \right] dx = \frac{1}{2} \left[x - \frac{a}{6\pi} \sin\left(\frac{6\pi x}{a}\right) \right]_0^a = \frac{1}{2} \left[\left(a - \frac{a}{6\pi} \sin(6\pi) \right) - (0 - 0) \right] = \frac{a}{2}$$

②

$$\begin{aligned}\int_0^a \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{3\pi x}{a}\right) dx &= \int_0^a \frac{1}{2} \left[\cos\left(\frac{\pi x}{a} - \frac{3\pi x}{a}\right) - \cos\left(\frac{\pi x}{a} + \frac{3\pi x}{a}\right) \right] dx \\ &= \frac{1}{2} \int_0^a \left[\cos\left(\frac{2\pi x}{a}\right) - \cos\left(\frac{4\pi x}{a}\right) \right] dx = \frac{1}{2} \left[\frac{a}{2\pi} \sin\left(\frac{2\pi x}{a}\right) - \frac{a}{4\pi} \sin\left(\frac{4\pi x}{a}\right) \right]_0^a \\ &= 0\end{aligned}$$

so we're

$$\int_0^a |\psi(x)|^2 dx = \frac{2}{5a} \left[\left(\frac{a}{2}\right) + 4\left(\frac{a}{2}\right) + 0 \right] = \frac{2}{5a} \cdot \frac{5a}{2} = 1$$

and so ψ is normalized. Even though the second way involves more work, we'll need $|\psi(x)|^2$ for part (c) anyway, so good that we did the calculation.

(b) We've already said $C_n = 0$ for $n \neq 1$ or 3 . Since $|C_n|^2$ is the probability of measuring E_n as the energy, this tells us that we will never measure

E_2, E_4, E_5, E_6 or the like, and thus will only ever get $E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$

or $E_3 = \frac{9\pi^2 \hbar^2}{2ma^2}$. And since $|C_1|^2 = 1/5$ and $|C_3|^2 = 4/5$, this tells us that

we'll measure $\frac{\pi^2 \hbar^2}{2ma^2}$ 20% of the time and $\frac{9\pi^2 \hbar^2}{2ma^2}$ the other 80%.

(c) $|\psi(x)|^2$ is a probability density. Just like the total mass in a volume V is given by $\int_V \rho d^3\vec{r}$ if ρ is the mass density, $\int_V |\psi|^2 d^3\vec{r}$ is the probability that the particle described by ψ will be found in the volume V . Here, our "volume" is $a/2 < x < a$, the right half of the box. Thus, $\int_{a/2}^a |\psi(x)|^2 dx$ tells us the chance of seeing the particle there. $|\psi|^2$ was already obtained, and so we need three integrals to get the probability:

$$\begin{aligned}\int_{a/2}^a \sin^2\left(\frac{\pi x}{a}\right) dx &= \frac{1}{2} \left[x - \frac{a}{2\pi} \sin\left(\frac{2\pi x}{a}\right) \right]_{a/2}^a = \frac{1}{2} \left[\left(a - \frac{a}{2\pi} \sin(2\pi)\right) - \left(\frac{a}{2} - \frac{a}{2\pi} \sin(\pi)\right) \right] \\ &= a/4\end{aligned}$$

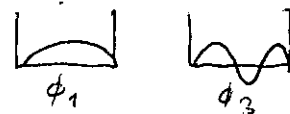
$$\begin{aligned}\int_{a/2}^a \sin^2\left(\frac{3\pi x}{a}\right) dx &= \frac{1}{2} \left[x - \frac{a}{6\pi} \sin\left(\frac{6\pi x}{a}\right) \right]_{a/2}^a = \frac{1}{2} \left[\left(a - \frac{a}{6\pi} \sin(6\pi)\right) - \left(\frac{a}{2} - \frac{a}{6\pi} \sin(3\pi)\right) \right] \\ &= a/4\end{aligned}$$

$$\int_{a/2}^a \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{3\pi x}{a}\right) dx = \frac{1}{2} \left[\frac{a}{2\pi} \sin\left(\frac{2\pi x}{a}\right) - \frac{a}{4\pi} \sin\left(\frac{4\pi x}{a}\right) \right]_{a/2}^a = 0$$

so the probability

$$P(a/2 < x < a) = \frac{2}{5a} \left[\left(\frac{a}{4}\right) + 4\left(\frac{a}{4}\right) \right] = \frac{2}{5a} \cdot \frac{5a}{4} = \frac{1}{2}$$

so half the time & lock, the particle will be in the right half. This is because both ϕ_1 and ϕ_3 are symmetric and the middle of the box:



Thus, ψ is as well as so $|\psi|^2$ is; thus, half of the area under the curve is on either side of $x = a/2$, and thus, the probability will be 50%.

(3)

P. 2

This problem deals with the properties of the energy eigenstates of the infinite square well, in particular, the expectation values and uncertainties. So, let's go!

(a) To find $\langle \hat{X} \rangle$, we simply need to use

$$\hat{X}\psi(t, x) = x\psi(t, x)$$

so that

$$\begin{aligned}\langle \hat{X} \rangle &= \int_{-\infty}^{\infty} \psi^*(t, x) (\hat{X}\psi(t, x)) dx \\ &= \int_{-\infty}^{\infty} \psi^*(t, x) x \psi(t, x) dx \\ &= \int_0^a \sqrt{\frac{2}{a}} e^{iENt/\hbar} \sin\left(\frac{N\pi x}{a}\right) \cdot x \cdot \sqrt{\frac{2}{a}} e^{-iENt/\hbar} \sin\left(\frac{N\pi x}{a}\right) dx \\ &= \frac{2}{a} \int_0^a x \sin^2\left(\frac{N\pi x}{a}\right) dx \\ &= \frac{1}{a} \int_0^a x \left[1 - \cos\left(\frac{2N\pi x}{a}\right)\right] dx \\ &= \frac{1}{a} \left[\frac{1}{2} x^2 - \frac{ax}{2N\pi} \sin\left(\frac{2N\pi x}{a}\right) - \frac{a^2}{4N^2\pi^2} \cos\left(\frac{2N\pi x}{a}\right) \right]_0^a \\ &= \frac{1}{a} \left[\frac{1}{2} a^2 - \frac{a^2}{2N\pi} \sin(2N\pi) - \frac{a^2}{4N^2\pi^2} (\cos(2N\pi) + 1) \right] \\ &= \boxed{\frac{a}{2}}\end{aligned}$$

since $\sin(2N\pi) = 0$ and $\cos(2N\pi) = 1$ for any integer N .

Plus, on average, the particle is in the middle of the box - not a huge surprise.

$\hat{P}$ acts on wavefunction as

$$\hat{P}\psi(t, x) = -i\hbar \frac{\partial \psi(t, x)}{\partial x}$$

so

$$\begin{aligned}\langle \hat{P} \rangle &= \int_{-\infty}^{\infty} \psi^*(t, x) \hat{P}\psi(t, x) dx \\ &= \int_0^a \sqrt{\frac{2}{a}} e^{iENt/\hbar} \sin\left(\frac{N\pi x}{a}\right) \left(-i\hbar \sqrt{\frac{2}{a}} \cdot \frac{N\pi}{a} \cos\left(\frac{N\pi x}{a}\right) \right) dx \\ &= -\frac{2N\pi\hbar i}{a^2} \int_0^a \sin\left(\frac{N\pi x}{a}\right) \cos\left(\frac{N\pi x}{a}\right) dx\end{aligned}$$

$$= -\frac{N\pi\hbar^2}{a^2} \int_0^a \sin\left(\frac{2N\pi x}{a}\right) dx = \frac{i\hbar^2}{2a} \cos\left(\frac{2N\pi x}{a}\right) \Big|_0^a$$

$$= \frac{i\hbar^2}{2a} [\cos(2N\pi) - 1] = 0$$

So, on average, the particle moves to the left just as much as it does to the right, and thus its average momentum is zero.

(b) Since $\hat{X}^2$ just multiplies a wavefunction by x^2 , we have

$$\langle \hat{X}^2 \rangle = \int_{-\infty}^{\infty} \psi^*(t, x) x^2 \psi(t, x) dx$$

$$= \frac{2}{a} \int_0^a x^2 \sin^2\left(\frac{N\pi x}{a}\right) dx$$

$$= \frac{1}{a} \int_0^a \left(x^2 - x^2 \cos\left(\frac{2N\pi x}{a}\right) \right) dx$$

$$= \frac{1}{a} \left[\frac{x^3}{3} - \frac{a x^2}{2N\pi} \sin\left(\frac{2N\pi x}{a}\right) - \frac{a^2 x}{N^2 \pi^2} \cos\left(\frac{2N\pi x}{a}\right) + \frac{a^3}{4N^3 \pi^3} \sin\left(\frac{2N\pi x}{a}\right) \right]_0^a$$

$$= \frac{1}{a} \left[\frac{a^3}{3} + \frac{a^3}{2N^2 \pi^2} (\cos(2N\pi) - 1) + \frac{a^3}{4N^3 \pi^3} \sin(2N\pi) \right]$$

$$= \boxed{\frac{a^2}{3} - \frac{a^2}{2N^2 \pi^2}}$$

$\langle \hat{P}^2 \rangle$ is easy to compute, because

$$\hat{P}^2 \psi = \left(-i\hbar \frac{\partial}{\partial x} \right)^2 \psi = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2}$$

$$= -\hbar^2 \frac{\partial^2}{\partial x^2} \left(\sqrt{\frac{2}{a}} e^{-\frac{iENt}{\hbar}} \sin\left(\frac{N\pi x}{a}\right) \right)$$

$$= -\hbar^2 \left(\sqrt{\frac{2}{a}} e^{-\frac{iENt}{\hbar}} \cdot -\frac{N^2 \pi^2}{a^2} \sin\left(\frac{N\pi x}{a}\right) \right)$$

$$= \frac{N^2 \pi^2 \hbar^2}{a^2} \psi$$

Therefore,

$$\langle \hat{P}^2 \rangle = \int_{-\infty}^{\infty} \psi^* (\hat{P}^2 \psi) dx$$

$$= \int_{-\infty}^{\infty} \psi^* \left(\frac{N^2 \pi^2 \hbar^2}{a^2} \psi \right) dx$$

$$= \frac{N^2 \pi^2 \hbar^2}{a^2} \int_{-\infty}^{\infty} |\psi|^2 dx = \boxed{\frac{N^2 \pi^2 \hbar^2}{a^2}}$$

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because $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$.

(c) We can now compute the uncertainties:

$$\begin{aligned}\Delta x^2 &= \langle x^2 \rangle - \langle x \rangle^2 \\ &= \frac{a^2}{3} - \frac{a^2}{2N^2\pi^2} - \frac{a^2}{4} \\ &= \frac{a^2}{12} - \frac{a^2}{2N^2\pi^2}\end{aligned}$$

or

$$\Delta x = \frac{a}{2} \sqrt{\frac{1}{3} - \frac{2}{N^2\pi^2}}$$

$\langle p \rangle = 0$, so

$$\Delta p = \sqrt{\langle p^2 \rangle} = \boxed{\frac{N\pi\hbar}{a}}$$

and thus

$$\begin{aligned}\Delta x \Delta p &= \frac{a}{2} \sqrt{\frac{1}{3} - \frac{2}{N^2\pi^2}} \cdot \frac{N\pi\hbar}{a} \\ &= \frac{\hbar}{2} \sqrt{(N\pi)^2 \left(\frac{1}{3} - \frac{2}{N^2\pi^2} \right)} \\ &= \frac{\hbar}{2} \sqrt{\frac{\pi^2 N^2}{3} - 2}\end{aligned}$$

I write it this way because it's easier to compare this with $\hbar/2$:

note that (i) $N \geq 1$; (ii) $\pi/3 > 1$; and (iii) $\pi > 3$.

Thus $\frac{\pi}{3} \cdot \pi \cdot N^2 > 1 \cdot 3 \cdot 1$, or $\frac{\pi^2 N^2}{3} > 3$. Thus,

$\frac{\pi^2 N^2}{3} - 2 > 1$, or thus $\sqrt{\frac{\pi^2 N^2}{3} - 2} > 1$. Now this confirms

that $\boxed{\Delta x \Delta p > \hbar/2}$, as it must be. (In fact, the ground state wavefunction with $N=1$ has the least value of $\Delta x \Delta p$, $\frac{\hbar}{2} \sqrt{\frac{\pi^2}{3} - 2} \approx 1.136 \cdot \hbar/2$.)

P. 3

This problem looks at a particle in a box that isn't in an energy eigenstate, namely,

$$\psi(x) = \begin{cases} 0 & x \leq 0, x \geq a \\ Cx(a-x) & 0 < x < a \end{cases}$$

(Note that $\psi(0) = \psi(a) = 0$, which is required by continuity of ψ . Thus, this is indeed a possible state of the system.)

(6)

(a) Obviously, $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$, so since $\psi = 0$ outside the box,

$$\begin{aligned} \int_{-\infty}^{\infty} |\psi(x)|^2 dx &= \int_0^a C^2 x^2 (a-x)^2 dx = C^2 \int_0^a (a^2 x^2 - 2a x^3 + x^4) dx \\ &= C^2 \left(\frac{a^2 x^3}{3} - \frac{2a x^4}{4} + \frac{x^5}{5} \right) \Big|_0^a = C^2 \left(\frac{a^5}{3} - \frac{a^5}{2} + \frac{a^5}{5} \right) \\ &= \frac{C^2 a^5}{30} \end{aligned}$$

so, assuming C is real and positive, $C = \sqrt{\frac{30}{a^5}}$

(b) Computing all these expectation values is easy, because ψ and all its derivatives are simply polynomials, whose integrals are piece-easy. So let's go:

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{\infty} \psi^*(x) x \psi(x) dx \\ &= \int_0^a \sqrt{\frac{30}{a^5}} x (a-x) \cdot x \cdot \sqrt{\frac{30}{a^5}} x (a-x) dx \\ &= \frac{30}{a^5} \int_0^a x^3 (a-x)^2 dx. \end{aligned}$$

Now, I could write this out and compute it, but notice that: let $y = \frac{a}{2} - x$ be a new variable of integration. Then

$$\begin{aligned} \langle x \rangle &= \frac{30}{a^5} \int_{a/2}^{-a/2} \left(\frac{a}{2} - y \right)^3 \left(\frac{a}{2} + y \right)^2 (-dy) \\ &= \frac{30}{a^5} \int_{-a/2}^{a/2} \left(\frac{a^2}{4} - y^2 \right)^2 \left(\frac{a}{2} - y \right) dy \end{aligned}$$

$$= \frac{30}{a^5} \int_{-a/2}^{a/2} \frac{a}{2} \left(\frac{a^2}{4} - y^2 \right)^2 dy - \frac{30}{a^5} \int_{-a/2}^{a/2} y \left(\frac{a^2}{4} - y^2 \right)^2 dy.$$

Notice that the first integrand is an even function of y and the second is an odd function. Since the limits are from $-a/2$ to $a/2$, the second integral vanishes, and the first is just twice the integral from 0 to $a/2$, so

$$\begin{aligned} \langle x \rangle &= \frac{30}{a^5} \cdot 2 \int_0^{a/2} \frac{a}{2} \left(\frac{a^2}{4} - y^2 \right)^2 dy = \frac{30}{a^4} \int_0^{a/2} \left(\frac{a^4}{16} - \frac{a^2 y^2}{2} + y^4 \right) dy \\ &= \frac{30}{a^4} \left(\frac{a^4}{16} y - \frac{a^2 y^3}{6} + \frac{y^5}{5} \right) \Big|_0^{a/2} = \boxed{\frac{a}{2}} \end{aligned}$$

which also makes sense: the wavefunction is symmetric about $x = a/2$, and all such wavefunctions will have $\langle x \rangle = a/2$.

(7)

$\langle \hat{x}^2 \rangle$, however, is a bit less intuitive:

$$\langle \hat{x}^2 \rangle = \int_{-\infty}^{\infty} \psi^*(x) \hat{x}^2 \psi(x) dx = \frac{30}{a^5} \int_0^a x^4 (a-x)^2 dx$$

$$= \frac{30}{a^5} \int_0^a (a^2 x^4 - 2a x^5 + x^6) dx$$

$$= \frac{30}{a^5} \left(\frac{a^2 x^5}{5} - \frac{2a x^6}{6} + \frac{x^7}{7} \right) \Big|_0^a = \frac{30}{a^5} \left(\frac{a^7}{5} - \frac{a^7}{3} + \frac{a^7}{7} \right)$$

$$= \boxed{\frac{2a^2}{7}}$$

Since $P\psi = -i\hbar \frac{\partial}{\partial x} \left[\sqrt{\frac{30}{a^5}} x(a-x) \right] = -i\hbar \sqrt{\frac{30}{a^5}} (a-2x)$, we have

$$\langle P \rangle = \int_{-\infty}^{\infty} \psi^*(x) P\psi(x) dx = -\frac{30i\hbar}{a^5} \int_0^a x(a-2x)(a-x) dx$$

$$= -\frac{30i\hbar}{a^5} \int_0^a (a^2 x - 3ax^2 + 2x^3) dx$$

$$= -\frac{30i\hbar}{a^5} \left(\frac{a^2 x^2}{2} - ax^3 + \frac{x^4}{2} \right) \Big|_0^a = -\frac{30i\hbar}{a^5} \left(\frac{a^4}{2} - a^4 + \frac{a^4}{2} \right)$$

$$= \boxed{0}$$

$$P^2\psi = P(P\psi) = -i\hbar \frac{\partial}{\partial x} \left(-i\hbar \sqrt{\frac{30}{a^5}} (a-2x) \right) = 2\hbar^2 \sqrt{\frac{30}{a^5}}$$

so

$$\langle P^2 \rangle = \int_0^a \left(\sqrt{\frac{30}{a^5}} x(a-x) \right) (2\hbar^2 \sqrt{\frac{30}{a^5}}) dx$$

$$= \frac{60\hbar^2}{a^5} \int_0^a (ax - x^2) dx = \frac{60\hbar^2}{a^5} \left(\frac{ax^2}{2} - \frac{x^3}{3} \right) \Big|_0^a = \boxed{\frac{10\hbar^2}{a^2}}$$

Now, we find $\langle P^2 \rangle$; let $H = \frac{P^2}{2m}$ here, so $\langle H \rangle = \frac{\langle P^2 \rangle}{2m}$,

giving

$$\boxed{\langle H \rangle = \frac{5\hbar^2}{ma^2}}$$

(c) $\Delta x = \sqrt{\frac{a^2}{28} - \left(\frac{a}{2}\right)^2} = \frac{a}{\sqrt{28}}$ and $\Delta p = \sqrt{\frac{10\hbar^2}{a^2} - 0} = \frac{\hbar}{a} \sqrt{10}$,

so

$$\Delta x \Delta p = \hbar \sqrt{\frac{10}{28}} = \frac{\hbar}{2} \sqrt{\frac{10}{7}}, \text{ and since } \sqrt{\frac{10}{7}} \approx 1.195, \text{ we see}$$

that $\boxed{\Delta x \Delta p > \hbar/2}$ is indeed satisfied.

(8)

(d) To determine the probability for us set E_3 , we calculate coefficient

c_3 in the expansion

$$\psi(x) = \sum_{n=1}^{\infty} c_n u_n(x) = \sqrt{\frac{2}{a}} \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{a}\right)$$

We know that with orthonormal basis, the inner product of ψ with u_n :

$$c_n = \langle u_n | \psi \rangle. \text{ Although we are not } c_3, \text{ let's find } c_n$$

for coefficients:

$$c_n = \langle u_n | \psi \rangle = \int_{-\infty}^{\infty} u_n^*(x) \psi(x) dx$$

$$= \frac{\sqrt{60}}{a^3} \int_0^a \sin\left(\frac{n\pi x}{a}\right) x(a-x) dx$$

$$= \frac{\sqrt{60}}{a^3} \left[\int_0^a ax \sin\left(\frac{n\pi x}{a}\right) dx - \int_0^a x^2 \sin\left(\frac{n\pi x}{a}\right) dx \right]$$

$$= \frac{\sqrt{60}}{a^3} \left[-\frac{a^2 x}{n\pi} \cos\left(\frac{n\pi x}{a}\right) + \frac{a^3}{n^2 \pi^2} \sin\left(\frac{n\pi x}{a}\right) + \frac{ax^2}{n\pi} \cos\left(\frac{n\pi x}{a}\right) - \frac{2a^2 x}{n^2 \pi^2} \sin\left(\frac{n\pi x}{a}\right) - \frac{2a^3}{n^3 \pi^3} \cos\left(\frac{n\pi x}{a}\right) \right]_0^a$$

$$= \frac{\sqrt{60}}{a^3} \left[-\frac{a^3}{n\pi} \cos(n\pi) + \frac{a^3}{n^2 \pi^2} \sin(n\pi) + \frac{a^3}{n\pi} \cos(n\pi) - \frac{2a^3}{n^2 \pi^2} \sin(n\pi) - \frac{2a^3}{n^3 \pi^3} \cos(n\pi) + \frac{2a^3}{n^3 \pi^3} \right]$$

$$\cos(n\pi) = (-1)^n \text{ and } \sin(n\pi) = 0, \text{ so}$$

$$c_n = \frac{2\sqrt{60}}{n^3 \pi^3} [1 - (-1)^n]$$

Note if n is odd, $(-1)^n = -1$ and if n is even, $(-1)^n = 1$, so

$$c_n = \begin{cases} \frac{4\sqrt{60}}{n^3 \pi^3} & n=1, 3, 5, \dots \\ 0 & n=2, 4, 6, \dots \end{cases}$$

$$c_3 = \frac{4\sqrt{60}}{27\pi^3}, \text{ so the chance of measuring } E_3 = \frac{q\pi^2 \hbar^2}{2ma^2} \text{ is}$$

$$|c_3|^2 = \left| \frac{4\sqrt{60}}{27\pi^3} \right|^2 = \frac{320}{729\pi^6} = 4.57 \times 10^{-4} = 0.0457\%$$

so, not very likely.